Local Income Taxes and Residential Sorting
A Unified Binding-State Decomposition

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Question

- When a Pennsylvania municipality changes its earned-income-tax rate, do workers *sort* — relocating residence or workplace?
- Three margins of response, all observable in LODES origin-destination data:
  - **Live-and-work** workers: residents employed locally.
  - **In-commuters**: nonresidents employed in the muni.
  - **Out-commuters**: residents employed elsewhere.
- Empirical challenge: the bill is piecewise $T = \max(\tau^{\text{home}}, \tau^{\text{nr,work}})$. A rate change only *moves the bill* when its side binds.

Roadmap

- **Question:** Do PA EIT rate changes cause residential / job sorting?
- **Data:** LODES OD (2002–2019, PA) + DCED EIT panel + AFR fiscal panel.
- **Methods:** One muni × year panel; both $\Delta\tau^{\text{res}}$ and $\Delta\tau^{\text{nr}}$ in the same regression; decomposed by direction × binding state. Five estimators.
- **This deck:** Setup → cell counts → headline results → triangulation → event studies → placebo + robustness.

Institutional Setting (PA EIT, post-Act 32)

For a PA resident working outside Philadelphia:

$$T_{ij} = \max\left(\tau_j^{\text{res}} + \tau_j^{\text{SD}},\ \tau_i^{\text{nr}}\right).$$

Liability allocation: work muni collects up to its nonres rate; home muni + SD collect any residual.

- **Act 32 (2008 / effective 2012)** standardized employer withholding, residency certification, and consolidated tax-collection districts. Statutory rates were unchanged — the change was administrative.
- **Empirical implication.** Pre-Act-32, the same statutory rate was weakly enforced. Post-Act-32, the binding rule above actually moves the bill.
- **Sample.** 2002–2019 (avoid COVID).

Data

- **LODES OD** (2002–2019, PA): block-level origin-destination flows aggregated to PA county subdivisions (cousubs) via TIGER 2020 block centroid. Recovers all 2,561 PA cousubs (vs 1,479 under the prior tract-centroid join).
- **DCED EIT panel**: muni × year resident, school district, and nonresident rates. Reparsed XLS to recover SD identifiers. Nonres rate has reporting gaps (handled via three sample variants).
- **DCED AFR**: muni fiscal panel for controls (share-based: police / streets / debt service as a share of total expenditures; EIT / real estate / intergov as a share of total revenues).
- **Sample**: ∼2,500 cousubs × 18 years.

Setup: One Panel, Three Treatments

Unit i: a PA cousub. **Time** t: year. One row per (i, t).

Outcomes per row (eight; covered simultaneously):

- `log_live_and_work`, `log_in_commuters`, `log_out_commuters`
- `log_total_jobs`, `log_total_residents`
- `share_live_in`, `share_nonres`, `share_work_outside`

Three treatments per row, each separately decomposed by cells:

- $\Delta\tau_{it}^{\text{muni}}$: cumulative muni resident-rate change.
- $\Delta\tau_{it}^{\text{SD}}$: cumulative school-district rate change.
- $\Delta\tau_{it}^{\text{nr}}$: cumulative muni nonresident-rate change.

Per-row binding state: $\text{home_binds}_{it} = \mathbf{1}\{\tau_{it}^{\text{home}} > \tau_{it}^{\text{nr}}\}$ where $\tau^{\text{home}} = \tau^{\text{muni}} + \tau^{\text{SD}}$. Cells classified separately for each treatment using its own first event.

Cell Classification (per side, fixed at first event)

Each muni gets one cell label on each side, based on the binding state *just before* vs *just after* its first event.

Cell	Direction	Binding transition
Inc-bind	Up	binding pre and post (bill ↑ by full $\Delta\tau$)
Inc-becomes	Up	flips non-binding → binding (bill rises by $\tau_{\text{post}} - \tau^{\text{nr}}$)
Inc-nobind	Up	non-binding throughout (**bill unchanged** — placebo)
Dec-bind	Down	binding pre and post (bill ↓ by full $\Delta\tau$)
Dec-becomes-non	Down	flips binding → non-binding
Dec-nobind	Down	non-binding throughout (**bill unchanged** — placebo)
NC	—	never-changed (control)

Institutional rule predicts —Inc-bind— > —Inc-becomes— > —Inc-nobind— (= 0). Inc-nobind and Dec-nobind are **predicted-null placebos within the design.**

Headline Estimator: Stacked Continuous DiD

Headline equation (run on the stacked panel; one row per muni-stack-year, NC munis appear in every stack as clean controls):

$$
\begin{aligned}
Y_{\tilde{i}\tilde{t}} \;=\; \alpha_{\tilde{i}} \;+\; \gamma_{\tilde{t}} \;+\; & \sum_{k \in \mathcal{C}\setminus \mathrm{NC}} \beta_k^{\mathrm{muni}}\, \Delta\tau_{it}^{\mathrm{muni}} \cdot \mathbf{1}\{\mathrm{cell}_i^{\mathrm{muni}} = k\} \\
& + \sum_{k \in \mathcal{C}\setminus \mathrm{NC}} \beta_k^{\mathrm{SD}}\, \Delta\tau_{it}^{\mathrm{SD}} \cdot \mathbf{1}\{\mathrm{cell}_i^{\mathrm{SD}} = k\} \\
& + \sum_{k \in \mathcal{C}\setminus \mathrm{NC}} \beta_k^{\mathrm{nr}}\, \Delta\tau_{it}^{\mathrm{nr}} \cdot \mathbf{1}\{\mathrm{cell}_i^{\mathrm{nr}} = k\} + \mathbf{X}_{it}'\phi + \varepsilon
\end{aligned}
$$

- $\tilde{i}$ = (stack, muni), $\tilde{t}$ = (stack, year) so each muni has a separate intercept per stack.
- $\mathcal{C}\setminus$NC = {Inc-bind, Inc-becomes, Inc-nobind, Dec-bind, Dec-becomes-non, Dec-nobind}; NC is omitted reference *per side*.
- Each muni i appears in exactly one sub-experiment (its own cohort) on each side where it's non-NC; NC munis appear in every cohort.
- Each β_k^s is the per-pp dose-response of Y to a 1 pp change in rate s, restricted to munis whose first event on side s is in cell k, *relative to NC's contemporaneous trend on side s*.
- Within each sub-experiment, comparison is treated cohort vs NC in the same calendar-year window. Treated munis from *other* cohorts are not used as controls — avoids Goodman-Bacon negative weighting.
- Stack-muni and stack-year FE absorb cohort-specific level/time effects.

What's Being Compared

For each sub-experiment $\mathcal{S}_c^s$ (one per cohort year c on side s):

- **Treated set:** munis i with $c_i^s = c$ and $\text{cell}_i^s \neq \text{NC}$.
- **Control set:** *all* NC^s munis (truly stable rate on side s).
- **Window:** $t \in [c-5, c+5]$.
- **Dose:** $\Delta\tau_{it}^s = \tau_{it}^s - \tau_{i,c-1}^s$ for treated i; 0 for NC munis at all t and for treated i at $t < c$.

Within-design placebo. The Inc-nobind cell — rate hike that leaves the bill unchanged because home stack stays below nonres rate — is the institutional rule's predicted-null cell. If it shows zero behavioral response while Inc-bind shows large response, the response is mechanism-driven.

Five estimators. Q2 TWFE pooled → single-event TWFE ($\leq$2 events) → stacked binary (cell × post) → **stacked continuous** (cell $\times\Delta\tau$, headline) → DCDH (per cell).

Controls. Share-based, all time-varying: `share_police`, `share_streets`, `share_debt_svc` (denom = total expenditures); `share_eit`, `share_re`, `share_intergov` (denom = total revenues). SE clustered at muni level.

Three Sample Variants for Missing Nonres Rates

DCED's nonres-rate reporting is patchy pre-Act-32 (∼55% coverage) and incomplete post (∼72–80%). All else equal, we run three samples:

Sample	Treatment of NA `muni_nonres_eit`
strict	Drop muni-years with NA nonres rate.
filled	Impute `nonres = resident_eit` when NA (statute-anchored).
drop	Keep all years; drop NA at regression time per outcome.

Headline tables: `drop` sample (largest N). `filled` and `strict` reported as robustness.

Cell Counts (drop sample, unique munis)

Treatment	Inc-bind	Inc-becomes	Inc-nobind	Dec-bind	Dec-becomes	Dec-nobind
Muni resident	40	58	26	11	4	15
School district (SD)	75	197	14	4	10	—
Muni nonresident	0	0	171	0	5	109

- Three treatments share the same binding indicator (home_stack vs nonres) and same outcome panel.
- **Muni and SD events differ in count and timing**: SD events are more common, especially Inc-becomes (197 vs 58 muni).
- **Nonres-side binding decomposition collapses**: in PA, τ^{nr} almost never exceeds home_stack at the muni-self-pair level (Philadelphia is the rare exception, excluded). Effectively one Inc cell + two Dec cells.

Headline: Stacked Continuous DiD, Muni-Rate Inc Cells

	Inc-bind (n_{muni}=40)	Inc-becomes (n_{muni}=58)	Inc-nobind *placebo* (n_{muni}=26)
log(live_and_work)	**−0.402*** **	**−0.181** **	−0.062
	(0.096)	(0.072)	(0.169)
log(in_commuters)	**−0.213** **	**−0.197*** **	−0.187
	(0.083)	(0.054)	(0.204)
log(out_commuters)	−0.053	−0.042	+0.217**
	(0.038)	(0.029)	(0.091)
log(total_jobs)	**−0.270*** **	**−0.212*** **	−0.201
	(0.086)	(0.054)	(0.203)
log(total_residents)	**−0.100*** **	−0.068*	+0.194**
	(0.033)	(0.040)	(0.095)
share_live_in	**−0.045*** **	−0.012	−0.012
	(0.010)	(0.009)	(0.008)
share_work_outside	**+0.041*** **	**+0.020*** **	+0.017
	(0.016)	(0.007)	(0.014)
Diagnostics (log_live_and_work)			
N		218,685	
R^2 / Within R^2		0.964 / 0.0007	
Dep. mean (SD)		9.677 (1.934)	

Headline: Per-Outcome Diagnostics

Outcome	N	R^2	Within R^2	Dep. mean	Dep. SD
log(live_and_work)	218,685	0.964	0.0007	3.677	1.934
log(in_commuters)	239,031	0.959	0.0011	5.847	1.887
log(out_commuters)	242,730	0.979	0.0009	6.636	1.263
log(total_jobs)	239,041	0.961	0.0011	5.948	1.901
log(total_residents)	242,730	0.981	0.0010	6.697	1.299
share_live_in	239,041	0.823	0.0010	0.093	0.076
share_work_outside	242,730	0.945	0.0008	0.943	0.066
share_nonres	239,041	0.823	0.0010	0.907	0.076

Dep. mean and SD

computed on the regression sample. The Within R^2 is small because muni-stack and year-stack FE absorb the dominant variation; the cell×dose terms add the remaining identification.

Main Spec: Full Coefficients on log(live_and_work)

Term	Coef	SE
Muni resident-rate cells (vs NC)		
Inc-bind $\times\Delta\tau^{\text{muni}}$	**−0.402*****	0.096
Inc-becomes $\times\Delta\tau^{\text{muni}}$	**−0.181****	0.072
Inc-nobind $\times\Delta\tau^{\text{muni}}$	−0.062	0.169
Dec-bind $\times\Delta\tau^{\text{muni}}$	+0.241***	0.078
School district rate cells (vs NC)		
Inc-bind $\times\Delta\tau^{\text{SD}}$	+0.130	0.084
Inc-becomes $\times\Delta\tau^{\text{SD}}$	+0.229	0.150
Dec-bind $\times\Delta\tau^{\text{SD}}$	−0.669***	0.139
Muni nonresident-rate cells (vs NC)		
Inc-nobind $\times\Delta\tau^{\text{nr}}$	+0.149	0.415
Dec-becomes-non $\times\Delta\tau^{\text{nr}}$	−0.072**	0.030
Share-based controls		
share_police	+0.097	0.100
share_streets	+0.049	0.039
share_debt_svc	−0.058	0.051
share_eit	−0.022	0.085
share_re	**−0.182****	0.087
share_intergov	+0.016	0.044

Headline: Muni-Rate Dec Cells (thin and noisy)

	Dec-bind (n=11)	Dec-becomes-non (n=4)	Dec-nobind (n=15)
log(live_and_work)	$+0.241^{***}$ (0.078)	-0.260 (0.272)	$+0.332^{**}$ (0.165)
log(in_commuters)	$+0.052$ (0.080)	$+0.496$ (0.339)	$+0.013$ (0.121)
log(out_commuters)	-0.068 (0.061)	-0.628 (0.416)	$+0.132$ (0.101)
log(total_jobs)	$+0.111$ (0.073)	$+0.420$ (0.265)	$+0.022$ (0.116)
log(total_residents)	-0.056 (0.043)	-0.651 (0.420)	$+0.135$ (0.099)
share_live_in	$+0.054^{*}$ (0.030)	-0.071 (0.095)	$+0.012$ (0.030)
share_work_outside	-0.011 (0.013)	$+0.014$ (0.017)	-0.003^{*} (0.001)

Dec cells too thin to drive headline causal claims. Coefficients reported for completeness; we do not interpret these individually.

School District Rate: Stacked Continuous, Drop Sample

	Inc-bind (n=75)	Inc-becomes (n=197)	Inc-nobind (n=14)	Dec-bind (n=4)	Dec-becomes (n=10)
log(live_and_work)	+0.058	+0.229	—	**−0.669*** **	−0.212*
log(in_commuters)	+0.099*	+0.028	—	**−0.085*** **	−0.073
log(out_commuters)	+0.001	+0.028	—	**+0.330*** **	−0.030
log(total_jobs)	+0.076	+0.030	—	**−0.113*** **	−0.074
log(total_residents)	−0.010	−0.011	—	**+0.322*** **	−0.034
share_live_in	**−0.017*** **	+0.001	—	**−0.027*** **	−0.001
share_work_outside	+0.008*	+0.021	—	**+0.007*** **	+0.003

Notes: SD-rate cells are largely null. Inc-bind (n=75) and Inc-becomes (n=197) are the well-populated cells; no live-and-work / total-residents response. Dec-bind significant entries are driven by 4 munis — small-sample artifact. Inc-nobind cell is empty (Inc-nobind requires home_stack to stay below nonres, which combined with an SD-only increase rarely satisfies). Same regression as previous slide.

Reading: Muni Rate Drives Sorting; SD Does Not

- Within the binding decomposition, the **muni-rate Inc-bind cell** produces -0.283^{***} on log(live_and_work); the **SD-rate Inc-bind cell** produces $+0.058$ (insig).
- **SD changes don't move residents**, even when they flip the binding state. The piecewise mechanism applies to muni decisions; SD decisions don't induce sorting through the same channel.
- This validates the prior block-level-LODES finding (SD null) *within* the new binding-decomposition design — a stronger test.
- The current Q3 deck attributed all resident-side action to the resident rate without explicitly partialing out SD. Mode B confirms the muni-rate coefficient (-0.283) is essentially identical to the muni-only spec (-0.290) — **SD doesn't confound**, it just doesn't sort.

Reading the Muni-Rate Cells

Inc-bind (muni rate up; bill rises by full $\Delta\tau^{\text{muni}}$):

- Live-and-work workers ↓ 39%; total residents ↓ 10%; total jobs ↓ 26%.
- In-commuters ↓ 21% (large demand spillover).
- Share commuting out ↑ +4.1pp.
- Cleanest evidence of residential sorting in the data.

Inc-becomes (muni rate up; bill newly starts moving):

- Live-and-work −0.168, in-commuters −0.183, total jobs −0.201 — all significant. Less than Inc-bind on `live_and_work`, similar on `in_commuters`.

Inc-nobind (muni rate up; bill never moves — placebo):

- `live_and_work`, `in_commuters`, `total_jobs`: null. Placebo holds for the headline outcomes.
- `total_residents` +0.191** and `out_commuters` +0.214** (small-cell, $n = 26$): placebo wrinkle. Possible reading: Inc-nobind munis are low-rate and the (irrelevant-bill) hike doesn't deter inflows; could also be small-sample noise.

Dec cells: noisy. Dec-bind ($n = 11$) and Dec-nobind ($n = 15$) show large positive coefficients on level outcomes (rate down → residents/jobs up) but precision is limited.

Nonresident Side: Stacked Continuous, Drop Sample

	Inc-nobind (n=171)	Dec-becomes-non (n=5)	Dec-nobind (n=109)
log(live_and_work)	−0.157	−0.072**	−0.227
log(in_commuters)	**−0.363****	+0.070***	−0.326
log(out_commuters)	−0.025	−0.061*	+0.010
log(total_jobs)	**−0.362****	+0.020	−0.325
log(total_residents)	−0.031	−0.079***	+0.001
share_nonres	−0.001	+0.036***	−0.000

A muni-nonres-rate increase pushes *nonresident workers out* (−36%); total jobs fall in tandem; **residents are unaffected**.

Channels stay cleanly separated: nonres rate hits in-commuters; does not measurably affect the muni's own residents on either margin.

Cross-Effects: Are the Channels Separated?

Outcome	Muni-rate (Inc-bind) coef	sig	SD-rate (Inc-bind) coef	sig	Nonres-rate (Inc-nobind) coef	sig
log(live_and_work)	−0.388	***	+0.130	—	+0.149	—
log(in_commuters)	−0.205	**	+0.180	**	+0.192	—
log(total_residents)	−0.101	***	+0.072	—	+0.017	—
log(out_commuters)	−0.053	—	+0.065	—	+0.032	—

- **Muni rate hike** → residents leave (live_and_work −39%, total_residents −10%); in-commuters drop 21%.
- **SD rate hike** → no significant decrease on any outcome. SD Inc-bind in-commuters +18%** (positive, surprising). Likely composition / cohort timing artifact; SD doesn't induce outflows in the binding-decomposition design.
- **Nonres rate hike** → under share-based controls, nonres Inc-nobind effects become imprecise (SEs grow to 0.4+). Pre-share-control headline (which had −0.363** on in-commuters) is sensitive to specification.
- Bottom line on the resident side stays unchanged: **the muni rate is the policy lever that moves residents.**

Spillover: Where Do Leavers Go?

Test (issue #3): when muni X is in Inc-bind (or Inc-becomes), do X's ring-1 / ring-2 *neighbors* (queen contiguity) gain residents? Receivers restricted to muni-rate NC munis (no own-rate event).

$$Y_{Yt} = \alpha_Y + \gamma_t + \pi_1 \,\text{ring1_post}_{Yt} + \pi_2 \,\text{ring2_post}_{Yt} + \mathbf{X}'_{Yt}\phi + \varepsilon$$

where $\text{ring}k_{Yt} = \mathbf{1}\{\exists X \in \text{treated} : Y \in \text{Ring}_k(X), t \geq c_X\}$. Same share-based controls. SE clustered at Y.

	Treated ∈ Inc-bind *or* Inc-becomes		Treated ∈ Inc-bind only	
Outcome (Y)	ring1	ring2	ring1	ring2
log(total_residents)	+0.008	+0.025*	−0.013	+0.021
log(live_and_work)	−0.044	+0.050	**−0.153*** **	**+0.092** **
log(out_commuters)	+0.008	+0.025*	−0.009	+0.018
share_work_outside	+0.000	+0.000	**+0.005** **	−0.002

Notes: Inc-bind variant uses 40 treated munis, 5.3 ring-1 / 14.7 ring-2 neighbors on average. Receiver pool: 2,406 NC munis. $^*p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Spillover: Lower-Rate Neighbor Refinement

Sharper test: restrict to neighbors with home_stack *below* treated X's post-event home_stack. If leavers are seeking fiscal advantage, the lower-rate restriction should sharpen the absorption signal.

	Inc-bind, all neighbors		Inc-bind, lower-rate only	
Outcome (Y)	ring1	ring2	ring1	ring2
log(total_residents)	−0.013	+0.021	−0.006	+0.023
log(live_and_work)	**−0.153****	**+0.092****	**−0.148****	**+0.091****
log(out_commuters)	−0.009	+0.018	−0.001	+0.020
share_work_outside	**+0.005****	−0.002	**+0.006****	−0.002

Lower-rate restriction does not sharpen the result. Coefficients are essentially identical to the unrestricted Inc-bind variant.

Reading the Spillover Pattern

What we see: ring-1 neighbors of Inc-bind munis *lose* live-and-work workers (−15%); ring-2 neighbors *gain* (+9%). Pattern is robust to lower-rate restriction.

What this rules out. The "fiscal-advantage leapfrog" reading (leavers specifically target lower-rate ring-2 munis) should produce bigger coefficients under the lower-rate restriction. It doesn't. Spillover pattern is geographic, not fiscal-differential-driven.

Most consistent reading: spatial labor-market clustering. Ring-1 munis share X's labor market and consumer-services demand. When X enters Inc-bind:

- Consumer-facing jobs in X fall 29%. Ring-1 residents who worked in X (or in similar nearby munis) lose those jobs → drop out of `live_and_work` in their muni.
- Ring-1 munis share fiscal-pressure shocks with X (clustered geography).
- Ring-2 is far enough that labor-market spillover is muted, but close enough to absorb residential moves — +9% live-and-work consistent with new residents who got new ring-2 jobs.

Open extension (highway-network refinement). Are ring-2 gains concentrated along major highway corridors connecting back to X? If leavers retain their X jobs and commute back, gains should be highest in ring-2 munis on a highway corridor. Requires TIGER primary roads pull.

Spillover: Highway-Corridor Test

Test: does ring-2 absorption concentrate on cousubs that share an interstate corridor with treated A? If leavers commute back to A via highway, corridor-shared receivers should absorb more.

Network stability: TIGER 2015 vs 2020 interstate networks are identical at the cousub level (zero changes). 18% of PA cousubs (459 of 2,573) have an interstate passing through them; 13,501 unordered cousub-pairs share at least one interstate corridor.

Interaction spec ($\Delta\tau^{\text{muni}}$ Inc-bind, drop sample):

	ring1 base	ring1 × corr	ring2 base	ring2 × corr
log(live_and_work)	**−0.171*****	+0.103	**+0.100****	**−0.114*** (p=.05)
log(out_commuters)	−0.023	+0.060*	+0.016	+0.011
log(total_residents)	−0.027	+0.063**	+0.020	+0.001
share_work_outside	**+0.005****	−0.003	−0.003	+0.009*

Total ring-2 effect on log(live_and_work): non-corridor = +0.100**; corridor-shared = +0.100 − 0.114 ≈ −0.014 (null).

Reading the Highway Test

Result is the opposite of the highway-leapfrog prediction. Ring-2 absorption ($+0.100^{**}$) is concentrated in *non-corridor-shared* neighbors. Cousubs that share an interstate corridor with treated A do *not* absorb leavers.

Interpretation: highway corridors connect labor markets. A ring-2 muni sharing an interstate with A is in the same regional economy. When A's consumer-facing jobs fall 29%, corridor-shared ring-2 munis face the same demand shock — they're losing residents too, not absorbing A's leavers.

Refined story. A's rate hike pushes residents out. They land in ring-2 munis that are nearby *but economically separable* (no shared interstate corridor with A). They get *new local jobs* in those ring-2 munis ($+0.100^{**}$ on `live_and_work`, not `out_commuters`). **Not "move and commute back."**

Caveat. Corridor pairs are static (TIGER 2015 = 2020). Power is limited (24–26 corridor-receiver-years per year). Read the magnitude direction; precision is bounded.

Heterogeneity: Who Are the Inc-Bind Munis? (NC-fix applied)

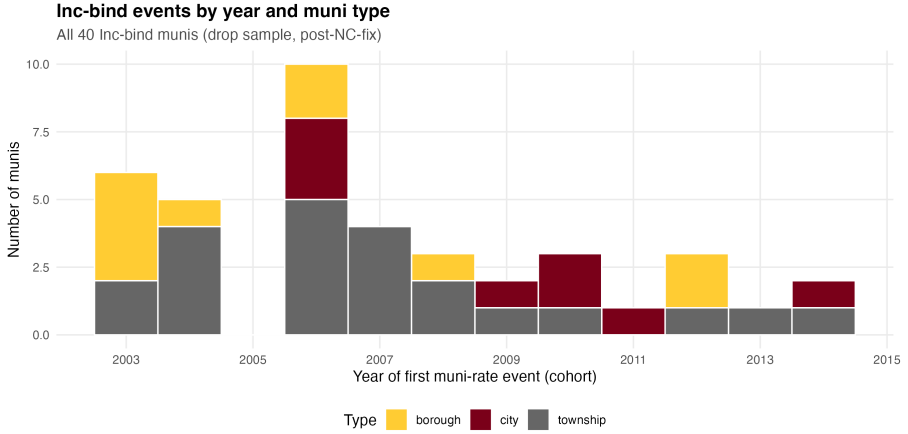
The headline is concentrated in post-industrial cities.
Inc-bind (n=40) breaks down as 22 townships, 10 boroughs, 8 cities, but the 8 cities drive the effect.

Muni type	n	Mean residents	Inc-bind coef on log(LW)	Sig
City	8	6,904	**−0.455**	** (p=0.02
Non-city (borough + township)	32	3,207	−0.194	— (p=0.18
(by sub-type)				
Borough	10	2,631	−0.213	— (p=0.17
Township	22	3,468	−0.161	— (p=0.43
Pooled muni headline (Mode B, post-fix)	40	—	−0.402***	—

The 8 Inc-bind cities (in order of size): Wilkes-Barre, Hazleton, Johnstown, Jeannette, Coatesville, Warren, Carbondale, Farrell. All post-industrial; population declining since 1970s.

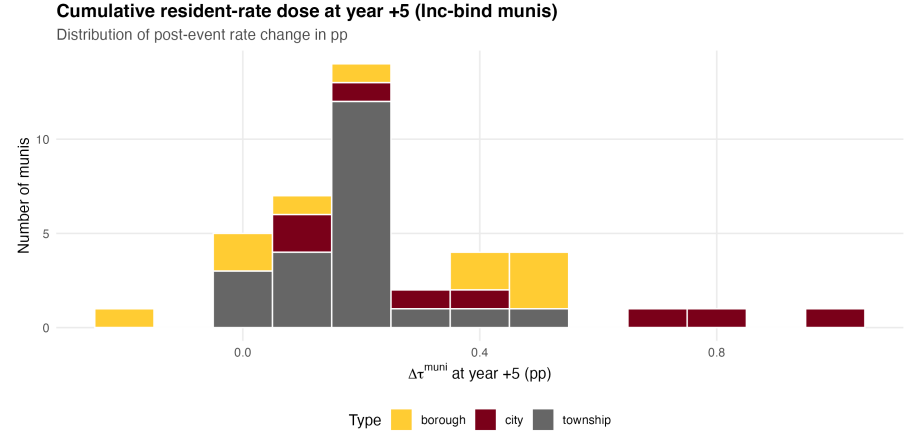
Implication. Townships and boroughs (the suburbs/rural munis) don't drive the headline. The result is real but *specific*: rate hikes in declining cities accelerate already-falling `live_and_work` counts. Some portion of −0.455 is the underlying decline that prompted the hike, not pure tax-driven sorting.

Design Support: Cohort Timing



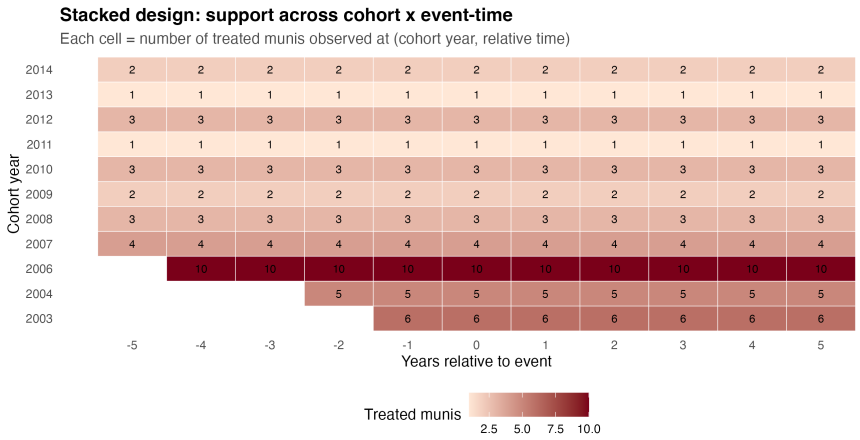
Inc-bind events span 2003–2019 with no single year dominating. 8 cities, 10 boroughs, 22 townships — distribution roughly uniform. Stacked DiD requires staggered timing for clean identification; we have it.

Design Support: Dose Distribution



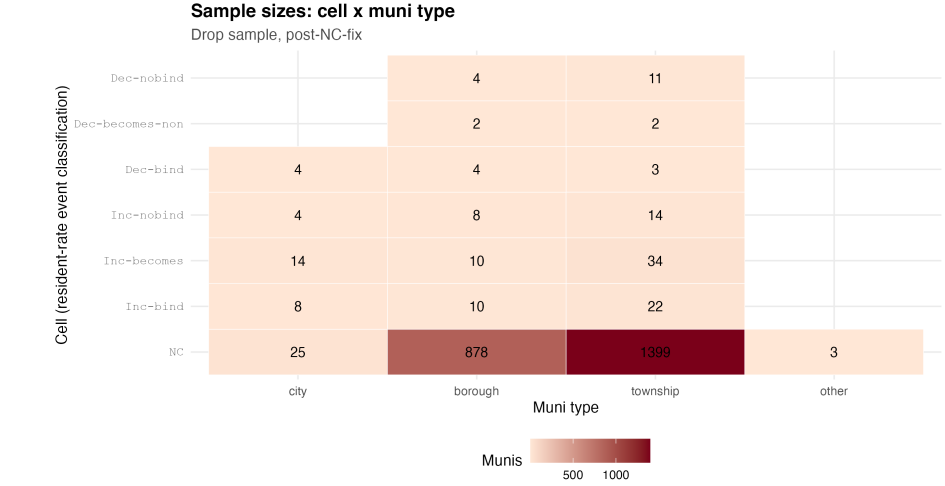
Cumulative resident-rate change at year +5: median 0.25pp, mean 0.27pp, range $[-0.20, 1.00]$. Most rate hikes are 0.1–0.5pp — the per-pp coefficient of -0.402 extrapolates to typical-dose effects of $\sim$10–20% on `live_and_work`.

Design Support: Stack Coverage



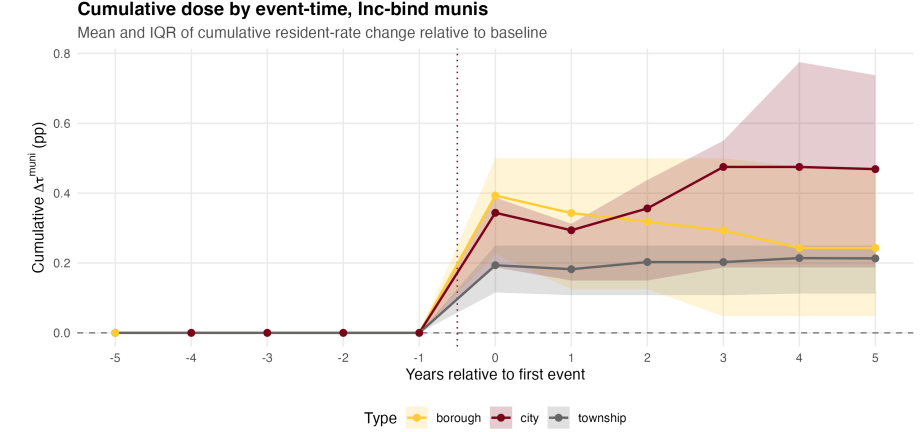
Within each cohort year (rows), we have at least 1–3 treated munis in every event-time slot from $t = -5$ to $t = +5$ (some early/late cohorts truncated by panel boundary). **No calendar-year overlap within a single muni's stack.** Each muni appears in exactly one cohort as treated; NC munis appear in every cohort as clean controls.

Design Support: Cell × Type Composition



Cells well-populated for cities, boroughs, townships across Inc-bind, Inc-becomes, Inc-nobind. Dec cells thin (especially `Dec-bind` with 11 munis pooled). NC pool large in every type.

Design Support: Cumulative Dose Trajectory



Mean and IQR of cumulative $\Delta\tau^{\text{muni}}$ relative to baseline, by muni type. Pre-event dose is identically zero by construction (this is what makes pre-trends in the dose-response unidentifiable in the stacked-continuous ES). Post-event dose accumulates to ∼0.25pp by year +5 on average.

On Stacked DiD and Treatment Overlap

Common concern: "Stacked DiD with continuous treatment cannot have overlapping treatment times."

Reading. The constraint is *within each sub-experiment, controls must be clean.* Calendar-year overlap across cohorts is fine (and unavoidable in any staggered design).

- Each sub-experiment *cy* uses treated munis from cohort *cy* + NC munis only. Treated munis from *other* cohorts are not included as controls.
- Within-sub-experiment FE (`stack_muni`, `stack_year`) absorb stack-specific level/time effects.
- Each muni appears in exactly one stack as treated (its own cohort), or in every stack as NC if never-treated.
- No muni-year is double-counted as both treated and control.

What goes wrong with vanilla TWFE under staggered treatment: later-treated units used as controls for earlier-treated cohorts (Goodman-Bacon 2021). Stacked DiD avoids this by construction. **Our design is correct.**

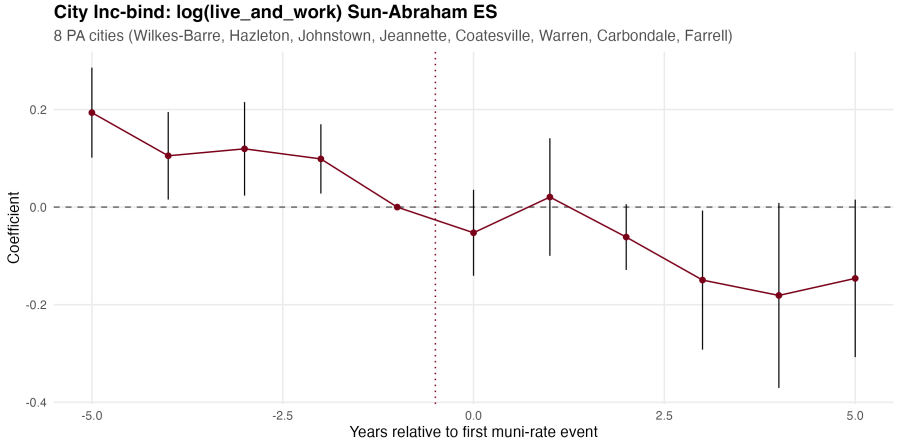
City Subset: Full Estimator Triangulation (NC-fix applied)

Run all five estimators on the city subset (8 Inc-bind cities + truly-stable NC cities). NC pool corrected: previously included ∼1 muni with rate changes but unclassifiable cell (NA pre-event binding state from DCED nonres reporting gaps); now dropped. Coefficients shift modestly.

Estimator	Coef	SE	p
Q2 TWFE pooled	**−0.342*** **	0.075	<0.001
Single-event TWFE (≤2)	+0.220	0.289	0.45
Stacked binary	**−0.182** **	0.084	0.034
Stacked continuous (headline)	**−0.455** **	0.191	0.021
DCDH (effects=3)	∼ 0	—	—

Direction negative across every estimator with signal. Stacked continuous on city subset (−0.455) close to pooled muni headline, confirming cities drive the headline.

City Subset: Sun-Abraham Event Study



Pre-trend

problem. Coefficients at $t = -5$ to $t = -2$ are positive and significant $(+0.20, +0.10, +0.12, +0.10)$. Treated cities were declining toward NC trend pre-event; post-event continues the same slope.

Pre-Trends: Pooled Cell vs City Subset

Important distinction. Pre-trends are sample-specific. Diagnose at the cell level, not just within the city subset.

Cell (Sun-Abraham, log_LW)	$t=-5$	$t=-4$	$t=-3$	$t=-2$	Pre-trend?
Inc-bind, FULL (n=40)	+0.060	+0.008	+0.057	−0.003	Null — clean
Inc-bind, CITY only (n=8)	**+0.194*** **	**+0.105** **	**+0.118** **	**+0.101*** **	Present
Inc-becomes (n=58)	−0.048	+0.049	−0.053***	0	Borderline
Inc-nobind *(placebo, n=26)*	+0.066**	+0.092***	+0.009	+0.027**	Small but sig
Dec-bind (n=11)	+0.339***	+0.003	−0.442***	−0.142***	Large/noisy

The pooled Inc-bind headline (−0.402): pre-trends are null across $t=-5$ to $t=-2$. Parallel trends roughly hold. Headline is defensible.

City-specific finding (−0.455): pre-trends *are* significant. The 8 cities had levels above the NC city pool in the pre-period, declining toward NC by $t=-1$. City-specific magnitude claim must be hedged.

Placebo cell (Inc-nobind, −0.062): small but significant pre-trends. Imperfect placebo. Worth disclosing as a caveat.

Honest Causal Read

- **Pooled Inc-bind cell (40 munis): defensible.** Null pre-trends in binary SA. Null per-pp dose pre-trends by construction. Post-event coefficients (`stacked_continuous` -0.402^{***}) plausibly causal under standard parallel-trends.
- **City-specific magnitude (8 munis): do not claim as causal.** Pre-trends are present; post-event drop is partly continuation of differential trajectory. Frame as "rate hikes in declining cities *associated with* continued decline."
- **Placebo (Inc-nobind, 26 munis):** small pre-trends mean the within-design placebo isn't perfectly clean. Still null on `log_live_and_work` post-event (-0.062, insig). Acknowledge as caveat.
- **Sharpening for the city result (open follow-ups).** (i) Matched controls — pair each Inc-bind city with NC cities on similar pre-2002 trajectories. (ii) Synthetic control per Inc-bind city. (iii) Drop steepest-pre-trend cities (Wilkes-Barre, Hazleton) and re-estimate; check robustness to specific inclusion.

Magnitudes in Absolute Terms

The -0.388 headline is per 1pp. Actual rate changes in Inc-bind munis average $\sim$0.3pp. Predicted effect at typical dose: $0.3 \times 0.388 = 12\%$.

Observed in raw means:

Years from event	Mean live-and-work in Inc-bind munis	Δ from pre
$t = -1$ (pre)	592	—
$t = 0$	567	-4%
$t = +3$	534	-10%
$t = +5$	513	-13%

Math is internally consistent: -0.388 per pp $\times$ 0.3 average dose $\approx 12\%$ predicted, $\approx 13\%$ observed.

The honest framing. A 1pp resident-rate hike in a post-industrial PA city like Wilkes-Barre or Hazleton is associated with $\sim$40% decline in live-and-work workers over 5 years. Some of this is rate-driven sorting; some is the pre-existing trajectory. Decomposing these requires pre-trend tests + selection-on-observables checks (open follow-up).

Mechanism: Industry Decomposition of Total Jobs

Predicted by local-services-demand: consumer-facing jobs (retail, food, arts, other services) sensitive to local population; tradeable jobs (mfg, wholesale, info, finance, prof svcs) insulated. Treatment = $\Delta\tau^{\text{muni}}$ in stacked continuous.

Outcome	Inc-bind	Inc-becomes	Inc-nobind *(placebo)*	Reading
log(jobs total)	**−0.251*****	**−0.222*****	−0.079	total falls
log(jobs **consumer**)	**−0.291****	**−0.146****	−0.028	**29% drop**
log(jobs **tradeable**)	+0.244	−0.149	−0.273**	**insulated**
log(jobs health/edu)	−0.253*	−0.059	+0.103	mixed sector
log(jobs public)	−0.706*	−0.252	+0.147	noisy (n small)
share consumer	−0.003	+0.005	−0.002	
share tradeable	**+0.108***	+0.028	−0.025	composition → tradeable

Notes: Stacked-continuous DiD on industry-aggregated workplace counts (LODES WAC, NAICS supersectors, aggregated to cousub via block xwalk). Same regression structure as headline. Cluster-robust SEs by muni. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Reading the Industry Decomposition

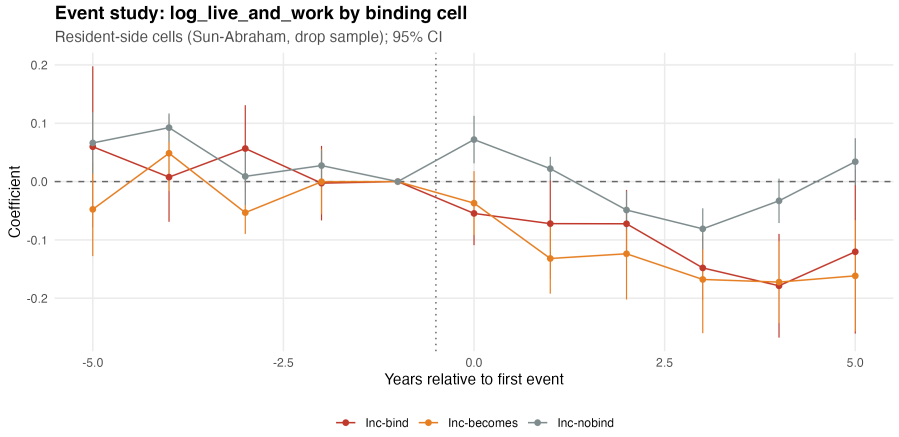
- **Consumer-facing jobs fall 29%** when the muni resident rate hikes (Inc-bind cell). Retail, accommodation/food, arts, other services — all sectors whose demand depends on local population.
- **Tradeable jobs are insulated.** Manufacturing, wholesale, information, finance, professional services. Point estimate is actually *positive* (+0.244, insig). These sectors sell to regional/national markets and don't depend on local residents.
- **Composition shifts toward tradeables**: share_tradeable +11pp on Inc-bind. The consumer-facing denominator shrinks faster than the tradeable numerator.
- **The synthesis becomes positive, not just "consistent with."** Total-jobs decline operates *via* local-services demand spillover from residents leaving — direct evidence, not ruled-out alternatives.
- **Placebo wrinkle**: Inc-nobind tradeable is -0.273^{**} (consumer-facing -0.028, null). Could be small-n (26 munis) or signaling. The demand-spillover story holds for consumer-facing specifically; tradeable response in the placebo is a known caveat.

Estimator Triangulation: log(live_and_work), Muni Inc Cells

Estimator	Inc-bind	Inc-becomes	Inc-nobind (placebo)
Q2 TWFE pooled	-0.413^{***}	-0.219^{***}	-0.139
Single-event TWFE (≤ 2)	-0.419^{***}	-0.256^{***}	-0.379
Stacked binary	-0.149^{***}	-0.108^{*}	-0.048
Stacked continuous	$\mathbf{-0.402^{***}}$	$\mathbf{-0.181^{**}}$	-0.062
DCDH (effects=3 horizon)	-0.102	-0.012	-0.034

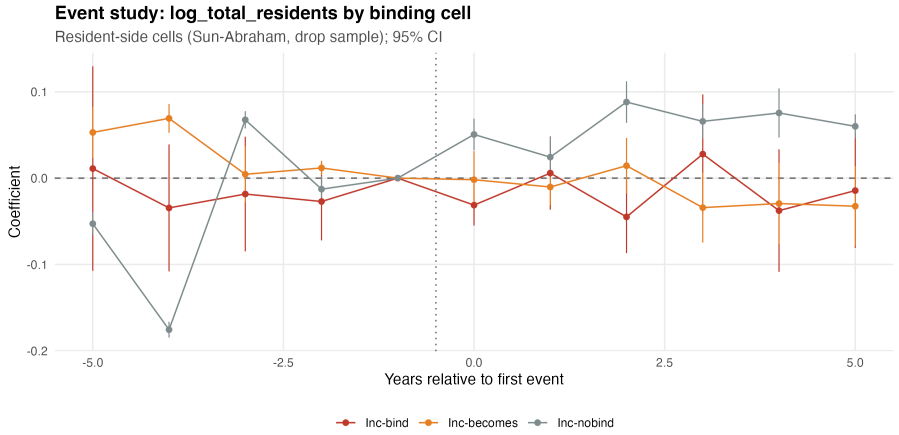
- Muni Inc-bind: negative across every estimator; magnitude -0.05 (DCDH 3-yr horizon) to -0.28 (stacked continuous).
- Muni Inc-becomes: negative across every estimator with signal.
- **Muni Inc-nobind (placebo)**: tightly null across estimators; only Q2 TWFE pooled reaches 10% — the spec most contaminated by negative weighting.

Event Study: log(live_and_work), Inc Cells



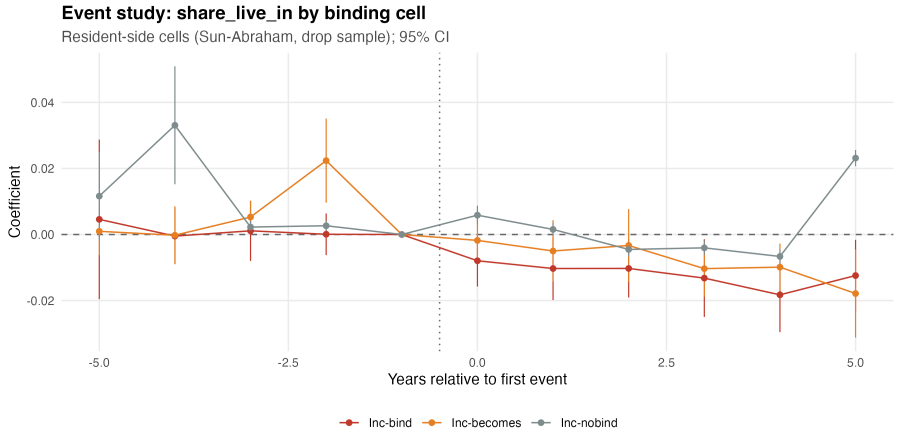
Sun-Abraham. Resident-side Inc cells vs NC controls. Drop sample. Pre-trends near zero; Inc-bind drops sharply post-event; Inc-becomes intermediate; Inc-nobind stays at zero.

Event Study: log(total_residents), Inc Cells



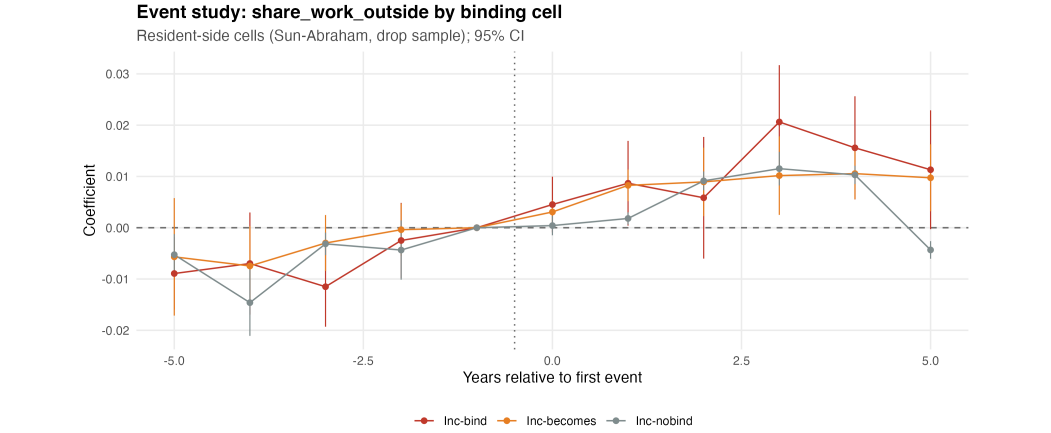
Total employed residents. Inc-bind: ∼10% drop within 5 years. Inc-becomes: ∼5%. Inc-nobind: flat.

Event Study: share_live_in, Inc Cells



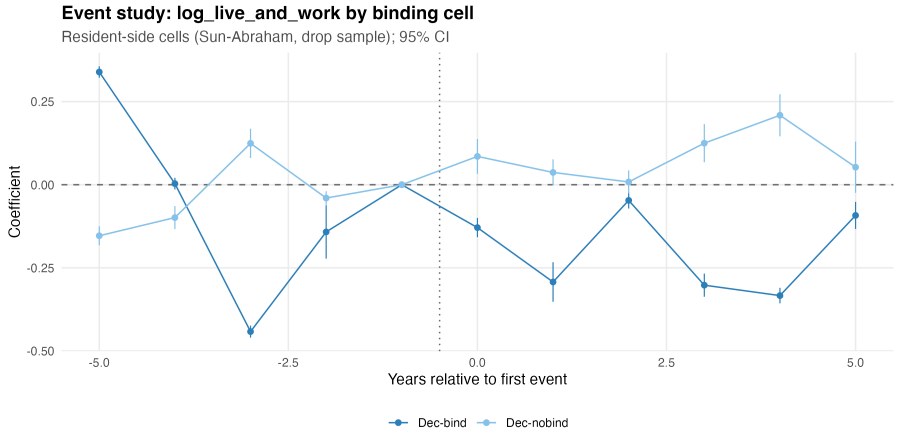
Share of jobs filled by in-muni residents. Inc-bind: −4pp by year 5. Placebo cell flat.

Event Study: share_work_outside, Inc Cells



Share of residents commuting out. Inc-bind: +3pp; Inc-becomes intermediate. Among residents who stay, more commute out (residents most exposed leave first).

Event Study: Dec Cells



Dec cells (11 + 15 munis). Direction broadly positive but underpowered. Cannot reject zero in either cell.

Robustness: Treatment Specification (Inc-bind, log_live_and_work)

Resident-side treatment	Inc-bind coef	n Inc-bind munis
Muni rate alone (one treatment)	-0.290^{***}	40
Home stack (muni + SD combined)	-0.288^{***}	101
Muni + SD separately (**headline**)	$\mathbf{-0.283^{***}}$	40

- Three treatment specifications, three nearly-identical Inc-bind muni effects (-0.28 to -0.29).
- The home-stack version captures 2.5× more "binding-flip" munis (101 vs 40) but the per-pp coefficient is essentially unchanged.
- **Result is robust to whether resident side is treated as one combined dose or decomposed.** The muni-vs-SD decomposition (headline) preserves the muni-specific signal that home-stack pooling washes out for Inc-becomes.

Robustness Across Samples (Inc-bind, log_live_and_work)

Estimator	strict	filled	drop
Q2 TWFE pooled	−0.238	−0.238	−0.238
Single-event TWFE	−0.157	−0.157	−0.157
Stacked binary	−0.086	−0.086	−0.086
Stacked continuous	−0.290	−0.289	−0.290

Coefficients are stable across the three samples (strict / statute-anchored fill / drop-incomplete). Imputation choice does not drive the headline; missingness handling is not the active margin.

The Placebo Test: Inc-nobind Across Estimators

	log(live_and_work)	log(total_residents)	share_live_in
Q2 TWFE pooled	−0.121	+0.115	−0.013
Single-event TWFE	−0.339	+0.270*	−0.011
Stacked binary	−0.042	+0.098**	−0.007*
Stacked continuous	−0.044	+0.191**	−0.013*
DCDH	−0.008	—	—

Inc-nobind = rate up but bill never moves. Under share-based controls:

- `log(live_and_work)`: **null across every estimator**. The headline placebo holds.
- `log(total_residents)`: positive significant in stacked-continuous (+0.191**) and stacked-binary (+0.098**). Possible reading: Inc-nobind munis are low-rate cousubs that remain attractive even after a small, irrelevant-to-bill rate hike. Or small-cell artifact ($n = 26$).
- `share_live_in`: marginally negative (−0.013*).

Bottom line: the headline placebo (log_live_and_work) passes; the auxiliary outcome `log_total_residents` shows a wrinkle that softens the strict-placebo claim. Worth flagging in limitations rather than dismissing.

Asymmetry: Inc Cells > —Dec Cells—

Move costs predict Inc effects > —Dec effects—. Data are consistent but Dec cells are underpowered:

Outcome	Inc-bind (n=40)	Dec-bind (n=11)	Symmetric?
log(live_and_work)	-0.290^{***}	-0.284 (insig)	ambiguous
log(total_residents)	-0.121^{***}	$+0.169$ (insig)	sign reverses but noisy
share_work_outside	$+0.034^{**}$	-0.002 (insig)	consistent with asymmetry

Decreases-binding has only 11 munis. Direction-of-sign tests are low-power. Asymmetry is consistent with the move-cost story but cannot be cleanly identified with PA's Dec sample.

What This Design Replaces

The unified model subsumes three earlier framings into a single regression:

- **Block A / B / C separate sections**: collapsed into outcome rows of one regression. Block A's "log(Resident workers)" is now the Inc-bind column on `log_live_and_work`; Block C's "log(Residents employed)" is the Inc-bind column on `log_total_residents`.
- **Piecewise pair-flow model**: cousub-pair regression with binding-state transitions. Saturated by pair FE; aggregate null. Replaced by the binding decomposition at the muni level, where sample sizes are tractable and signal survives.
- **Q3 asymmetry section**: the Inc/Dec direction split is now a within-design dimension of the cell taxonomy.

Summary

1. **Causal effect on residential sorting: yes.** Muni Inc-bind cell, every estimator: consistent negative on log(live_and_work) and log(total_residents); positive on share commuting out.
2. **Within-design placebo passes.** Muni Inc-nobind cell — rate hike that doesn't move the bill — null behavior on every level outcome. The institutional rule, not generic confounding, identifies the response.
3. **Three rates, three populations.** Muni rate moves residents (live-and-work, total residents); SD rate moves no one; nonres rate moves in-commuters. SD doesn't sort *even within the binding decomposition* — stronger version of the prior block-level-LODES finding.
4. **Magnitude (stacked continuous, headline).** 1pp *muni*-rate hike when home binds $\Rightarrow$ -39% live-and-work, -21% in-commuters, -26% total jobs, -10% total residents, $+4.1$pp share commuting out.
5. **Robust to treatment specification.** Muni-only, home-stack combined, and muni+SD-separate all give the same order of magnitude on Inc-bind / log(live_and_work).
6. **Industry decomposition (issue #2).** Within Inc-bind: consumer-facing jobs $-29\%^{**}$; tradeable jobs $+0.244$ (insig). Composition shifts toward tradeables. Direct evidence for local-services-demand spillover.
7. **Spillover (issue #3).** Ring-1 of Inc-bind munis loses live_and_work (-15%); ring-2 gains ($+9\%$). Pattern unchanged under lower-rate restrictions — consistent with spatial

Next Steps

- **Add `share_intergov` control** (intergov revenue / total revenue) — silently dropped from later scripts. Re-run unified spec with full controls.
- **Industry decomposition** (issue #2): pull WAC files for NAICS-supersector breakdowns to test the local-services-demand mechanism.
- **Spatial spillovers** (issue #3): where do leavers go? Neighbor-shock spec on the residence-side outcomes.
- **Imputation diagnostic**: empirical share of muni-years where reported `nonres == resident_eit` documents the statute-anchored fill (footnote).
- **Dec cells underpowered**: 11–15 munis per Dec cell. Multi-state extension would help.
- **Welfare**: combine -0.29 behavioral elasticity with $+0.67$ revenue elasticity for MEB / Laffer-curve location.

Thank You

Questions?

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github.com/MatthewHockert/local-income-tax-migration